

Dhruv Athaide

Cyber Analyst · Red Team & Mobile Security · Kotlin & Flutter · Published Researcher

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PROFESSIONAL SUMMARY

Cyber Analyst specializing in Red Team & Mobile Security with **2+ years** shipping production Android and Flutter systems, including **4 apps independently taken to Google Play**. Architects offensive Android toolkits aligned with MITRE ATT&CK and owns AOSP hardening end-to-end, mentoring **25+ developers** while leading teams of **60+**. **Published researcher** - co-author of a peer-reviewed IGI Global book chapter (©2026) on Android vulnerability assessment and mobile threat monitoring.

TECHNICAL SKILLS

Android Development: Kotlin, Java, Jetpack Compose, Flutter, Dart, Android SDK, Firebase, MVVM, Clean Architecture, Room, WorkManager, Coroutines & Flow, Retrofit, OkHttp, Hilt, Play Billing, Material Design, Gradle, REST APIs

Engineering Practices: OOP & Design Patterns, Data Structures & Algorithms, Multithreading, Memory Management, Unit & Integration Testing, Code Review, Agile Sprint Planning, Low-Level Design (LLD)

Security & Systems: Encrypted Storage, Biometric Auth, OWASP Mobile Top 10, Applied Cryptography, Android Internals (AOSP, SELinux, JNI/NDK)

Cloud & Tools: TypeScript, JavaScript, Node.js, Firebase (FCM, Firestore, Auth), Azure, Docker, Git, CI/CD, MongoDB

WORK EXPERIENCE

DeepCytes Cyber Labs UK Mumbai, Maharashtra

August 2025 - Present

Cyber Analyst - Red Team & Mobile Security

- Spearheaded red team engagements across 10+ client environments, architecting a modular Android offensive toolkit aligned with the MITRE ATT&CK Mobile framework, improving simulation fidelity by **40%**.
- Architected a covert Android telemetry platform with real time multi channel data streaming and sub 100ms latency, using STUN/TURN NAT traversal and custom signalling.
- Designed a production grade mobile threat intelligence engine, reducing permission analysis latency by **30%** through parallelised traversal and Kotlin Coroutines concurrency handling.
- Built a BLE anti stalking detection system, reducing false positives by **35%** across **10,000+ device samples** using RSSI modelling and spatial temporal correlation.

DeepCytes Cyber Labs UK Mumbai, Maharashtra

July 2024 - June 2025

Red Team & Mobile Dev Team Lead (Internship)

- Led a cross functional team of **60+** specialists across penetration testing, threat intelligence, and vulnerability research, lifting project completion rates **15%** and delivery timelines **25%**.
- Delivered **4 enterprise grade platforms**, including an LMS and SaaS product, lifting client engagement **35%**.
- Spearheaded a hardened AOSP distribution with custom kernel level modifications and hardware security module integration, reducing OS attack surface by **65%**.

Truboard Partners Mumbai, Maharashtra

June 2024 - August 2024

Production & Software Testing Intern

- Fine tuned a financial covenant detection ML model, improving business process efficiency by **10%**.
- Built a real time asset monitoring solution for ride sharing services, targeting a reduction in operational costs.

DeepCytes Cyber Labs UK Remote

January 2024 - June 2024

AI R&D Team Lead (Internship)

- Conducted applied research on AI's role in cybersecurity, identifying new integration points and improving organizational AI literacy.
- Led the Digital Twin project lifecycle from concept through implementation.

DeepCytes Cyber Labs UK Mumbai, Maharashtra

July 2023 - December 2023

Cyber Security Intern

- Contributed to cyber intelligence research alongside AI R&D and Deep Web teams, improving threat intelligence capabilities by **10%**.
- Conducted independent research into deep web threat actor behavior, informing organizational security research priorities.

KEY PROJECTS

Guardient - Mobile Privacy & Threat Intelligence App (Google Play)

Kotlin, Android SDK, C++ (NDK), MVVM, Clean Architecture

- Solely architected a modular Clean Architecture / MVVM codebase with a heuristic risk scoring engine evaluating **100+ installed apps** in real time.
- Optimized scan performance using concurrent caching, cutting scan latency **30%** while sustaining fluid 60fps dashboard rendering.
- Achieved **100%** malware and spyware detection across a 10 to 20 app test suite entirely on device, shipped solo to Google Play.

LockGuard+ - Intruder Detection & Secure Evidence Management App (Google Play)

Kotlin, CameraX, C++ (NDK/JNI), AES-256

- Personally engineered a headless CameraX capture pipeline, achieving **100%** alert reliability with sub **500ms** delivery latency.
- Built an automated, policy compliant evidence retention system guaranteeing storage limits are never breached.
- Solo shipped a subscription based security product end to end, owning architecture, billing integration, and Google Play release.

NoSurveil - BLE Anti-Stalking & Tracker Detection App (Google Play)

Kotlin, Android SDK, Room, Hilt, C++ (NDK), MVVM

- Solely designed an $O(N+M)$ temporal alignment algorithm correlating location and device signal histories, cutting false positives **35%** across **10,000+ device samples**.
- Built a continuous background BLE scanning engine using in memory batch processing, sustaining under **2% battery drain per hour**.
- Reduced local database I/O bottlenecks by deduplicating and throttling scan results before persistence, maintaining high data fidelity at low overhead.

Torch - Enterprise Remote Data Protection & Device Management Platform

Kotlin, Jetpack Compose, C++ (NDK/JNI), Node.js/Express, React, Firebase, MongoDB

- Personally engineered a cryptographic erasure engine rotating across 5 symmetric ciphers, shredding files in RAM before disk overwrite for forensic unrecoverability.
- Built a scalable fleet heartbeat service tracking online and offline status across a **100 device fleet**, with automatic offline state pruning.
- Designed a hierarchical RBAC system enabling org and branch admins to securely act on devices strictly within their assigned scope.

Cyber Investigator Battlegrounds (CIB) - Investigative Puzzle & Simulation Game (Google Play)

Flutter, Dart, Firebase, Dio

- Directed architecture of a reactive game state engine using precomputed $O(1)$ lookup structures, eliminating UI lag during rapid gameplay state changes.
- Guided a team building a cache first authentication layer with single flight token refresh, preventing race conditions across concurrent requests.
- Oversaw dynamic branching logic calculating case outcomes in real time based on the exact combination of evidence a player collects.

EDUCATION

Indian Institute of Technology Patna Patna, India

August 2026 - July 2028

M.S. in Computer Science and Data Analytics

Amity University Mumbai Mumbai, India

July 2022 - June 2025

Bachelor of Computer Applications (BCA) | CGPA: 9.69 / 10.0

St. Xavier's College Mumbai, India

August 2020 - June 2022

12th Standard, Science Stream

RESEARCH & PUBLICATIONS

Design of Comprehensive Mobile Device Vulnerability Checking Application

March 2026

Book Chapter, Ch. 6 · Advanced Cybersecurity for Threats Exploitation and Digital Risk · IGI Global Scientific Publishing

M. H. Devare[†], **D. J. Athaide**, A. M. Devare & V. Nagpal | Amity University Mumbai. Presents a comprehensive Android VAPT application design covering on-device vulnerability assessment, Wi-Fi security auditing, and real-time threat monitoring.

DOI: 10.4018/979-8-2600-0747-1.ch006 **ORCID:** orcid.org/0009-0000-3554-9798

CERTIFICATIONS

- Google Cloud Computing Foundations Certificate - Credly Verified
- Google Developer Student Club Web Developer Certificate